# Yaksh Butani

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## Education

**Western University** — Graduation: April 2030
Bachelor of Engineering Science — London, ON

## Experience

**Spirol** — Jun 2026 - Aug 2026
Jr Controls Engineer Intern — Windsor, ON

- Independently designed and programmed a pass/fail inspection cell on an **Epson C8 6-axis robot**, owning mechanical design, wiring diagrams, and PLC/robot control logic.
- Cut cycle time from **35s to 2s (2,000–4,000 parts/shift)** by programming pick, controlled-drop, sensor-based detection, and sort sequencing in Epson RC+, including tool offsets and point teaching.
- Converted AU-870 tolerance specs into a sensor-based detection method that catches **sub-millimetre** part sink.
- Integrated Keyence GC-1000 safety controllers and radar-based guarding into the RC700 system, meeting **CSA compliance** across production lines.
- Built an operator HMI with live part counts, fault display, logging, sort-type selection, and cycle start/stop control.

**University of Windsor** — Jun 2025 – Aug 2025
Research Assistant Intern — Windsor, ON

- Programmed a **Universal Robots arm** (URScript/PolyScope) with trajectory planning for collision-free pick-and-place on a moving conveyor.
- Designed a conveyor sorting line using a high-resolution camera, with grayscale conversion and thresholding to preprocess images.
- Trained a **TensorFlow vision model** to classify ripe vs. rotten tomatoes in real time, triggering arm motion as fruit entered the pick zone (96% accuracy).
- Synchronized conveyor speed, arm timing, and vision feedback, increasing throughput by **30%.**
- Debugged the vision pipeline through systematic testing, isolating failure points across camera input, preprocessing, and model output.

**FIRST Robotics Team 4903 & 1229** — Sept 2023 - Apr 2026
Programming Lead & Operator — Windsor, ON

- Programmed robot software in **C++**, including swerve drive with field-oriented control, vision and encoder-based turret PID loops, and a motion-profiled multi-joint arm.
- Built **10+ autonomous routines** using sensor fusion and trajectory planning, cutting cycle times year-over-year by **30%.**
- Developed a Limelight 4 and AprilTag **localization system** for field positioning and autonomous navigation.
- Tuned **PID gains** across drivetrain and arm subsystems through iterative field testing to balance speed and precision.
- **Led a team of 7 student programmers** through subsystem ownership and code reviews to maintain reliability under competition deadlines

## Projects

**Ocean Plastic Waste detection** | Python, Tensorflow, Html, CSS, JavaScript

- Developed and trained a **TensorFlow object detection model** on an open-source underwater plastic dataset, accurately identifying plastics of various sizes, shapes, and colors under challenging lighting conditions.
- Applied **data preprocessing and augmentation** (rotation, scaling, color correction) to improve model robustness for underwater imagery.
- Achieved **96% detection accuracy**, providing reliable **recognition of plastic waste**, with potential applications for **autonomous underwater vehicles (AUVs)** in environmental cleanup.
- Fully documented the **training, evaluation, and deployment workflow**, creating a r**eproducible pipeline** for future research and teaching purposes.

**3-DOF Drone Stabilization** | MATLAB, Simulink, PID Controllers

- Tuned PID loops to stabilize the drone across roll, pitch, and yaw axes, reducing overshoot and improving responsiveness during **multi-directional maneuvers**.
- Integrated real-time **MATLAB simulations** with onboard sensors, allowing live feedback control and **precise trajectory tracking** under variable load and environmental conditions.
- Monitored **sensor outputs** and **flight performance** to fine-tune control parameters, ensuring **precise positioning and stable maneuvers** under varying conditions.
- Analyzed sensor data and flight logs to **identify instability sources** and **adjust controller gains** accordingly

## Technical Skills

**Languages**: Python, Java, C++, Matlab, URScript, JavaScript, HTML, CSS, SPEL+, PLC programming, Modbus RS485

**Developer Tools / Platforms:** Git, Linux, VS Code, PyCharm, ROS2, IntelliJ, Unity, Arduino IDE, Jupyter, Raspberry Pi, Onshape, SolidWorks, SolidEdge

**Libraries and others:** React, Tailwind CSS, Yolo v8, PyTorch, TensorFlow, NumPy, Neural Networks, OpenCV, GSAP, Matplotlib